Work along a displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate work and identify when a force does no work.

SUCCESS CRITERIA - I can...

1. Identify the force component along displacement.
2. Use meters and newtons to obtain joules.
3. Distinguish effort from mechanical work on an object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “displacement” to explain your model or result. Sentence starters are optional.

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

parallel: along the same direction as a line or surface

01 Warm-up: bring back an earlier idea

A simplified deflection adds 0.3 m/s sideways for a lead time of 50 s.
Find the modeled shift and name an assumption.

Word help: deflection = a change in the direction or path of motion

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

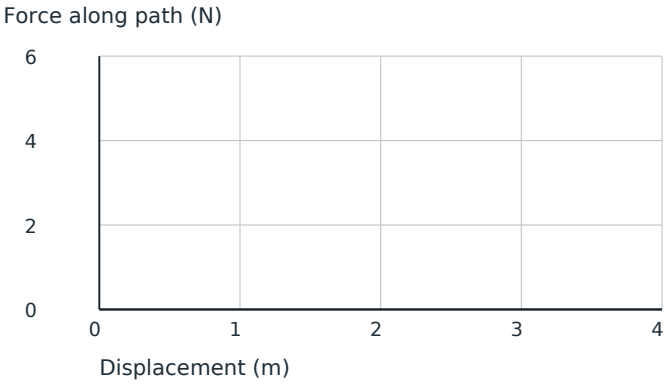
02 Worked example: explain the model

Reference: W = Fd.
Constant force parallel to displacement; W is work. No displacement means no work on the object by that force.

Worked reasoning: A constant 5 N pull acts along a 3 m displacement. Work is Fd=15 J. The object must move for this force to do mechanical work.

Your turn: Draw the force and displacement arrows.
Show the 15 J as area on a force-versus-position graph.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

Work along a displacement

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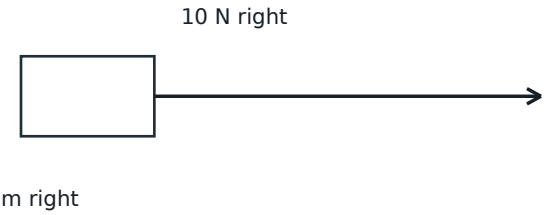
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A box moves 4 m right while a 10 N force acts right.
Draw force and displacement arrows; predict the sign of work.

Sentence starter: I predict … . My model supports this because … .



The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: 1 2 3

04 Apply the physics

You pull a box with the 10 N horizontal force shown in question 3.
Find the work over 4 m, then over 2 m.
Draw each force and displacement; explain why the shorter trip changes the work.

Sentence starter: My claim is … . The diagram/data show …, so … .

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student holds a heavy bag still. Is the upward support force doing mechanical work on the bag during the hold?
Explain without denying the student feels tired.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

Work along a displacement

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Continue down each column.

Sentence starters are optional. You may plan in any language.

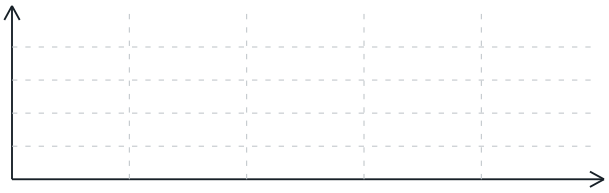
06 Review: connect an earlier idea

The same object stops from the same speed, but the stopping time is 4 times the original.
Compare average stopping-force magnitudes.
Show the signed impulse on the force-time model.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: Both … . They differ in … because … .

Force (N)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A 3 N force acts forward through 5 m.
Find its work.

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design two force-and-distance pairs doing 20 J of positive work.
Sketch both.

Sentence starter: My design shows … . It meets the requirement by … .

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3